Well, this is certainly progress, but for a long time scientists scratched their heads, wondering how to account for the remaining 70 percent of the universe, which, if inflationary cosmology was correct, had apparently gone AWOL. But then, in 1998, two groups of astronomers came to the same shocking conclusion, which brings our story full circle and once again reveals the prescience of Albert Einstein.

The Runaway Universe

Just as you may seek a second opinion to corroborate a medical diagnosis, physicists, too, seek second opinions when they come upon data or theories that point toward puzzling results. Of these second opinions, the most convincing are those that reach the same conclusion from a point of view that differs sharply from the original analysis. When the arrows of explanation converge on one spot from different angles, there's a good chance that they're pointing at the scientific bull's-eye. Naturally then, with inflationary cosmology strongly suggesting something totally bizarre—that 70 percent of the universe's mass/energy has yet to be measured or identified—physicists have yearned for independent confirmation. It has long been realized that measurement of the *deceleration parameter* would do the trick.

Since just after the initial inflationary burst, ordinary attractive gravity has been slowing the expansion of space. The rate at which this slowing occurs is called the deceleration parameter. A precise measurement of the parameter would provide independent insight into the total amount of matter in the universe: more matter, whether or not it gives off light, implies a greater gravitational pull and hence a more pronounced slowing of spatial expansion.

For many decades, astronomers have been trying to measure the deceleration of the universe, but although doing so is straightforward in principle, it's a challenge in practice. When we observe distant heavenly bodies such as galaxies or quasars, we are seeing them as they were a long time ago: the farther away they are, the farther back in time we are looking. So, if we could measure how fast they were receding from us, we'd have a measure of how fast the universe was expanding in the distant past. Moreover, if we could carry out such measurements for